12 th HACKER JUGAAD NOTES

ELECTROSTATIC POTENTIAL & CAPACITANCE

ELECTRIC POTENTIAL :-

- It Determine the Direction of flow of charge between two charge Bodies placed in contact with Each other.

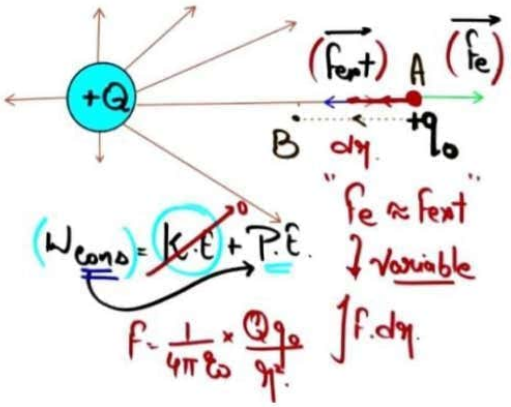
✡ The charge flows from H.P → L.P. The charge flow (stops) when potential becomes same.

ELECTROSTATIC POTENTIAL ENERGY

- The work done by Fext is negative of W.D by electric force and get fully stored in the form of potential Energy.

$$W_{AB} = \int_A^B F_{ext} \cdot dr = -\int_A^B F_e \cdot dr$$

Potential Energy.



- This work done against electrostatic get stored as potential Energy.

$$\Delta U = U_B - U_A = W_{AB}$$

✡ if nothing is mention consider only Fext (W.D)

- Electrostatic Potential Energy Difference between two points B and A is equal to minimum work done by external force in moving a test charge from A to B without Accn → $\Delta V = 0$

✡ The work done by an Electric field in moving a test charge. Depends upon initial and final Position not on Path.

F_e = conservative force
- state Function ✓
- Path Function ✗

Reference is required $U_{reff} = 0$

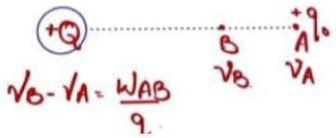
- Potential Energy at infinity zero.

$U_{reff} = 0$
U_∞ = zero

$\Delta U = U_B - U_A$

$\Delta U = U_B = W_{AB}$

ELECTROSTATIC potential



$V_B - V_A = \frac{W_{AB}}{q_0}$

If V_A and V_B are the Electrostatic potential.

$$V_B - V_A = \frac{W_{AB}}{q} = \frac{U_B - U_A}{q}$$

$$V_B - V_A = \frac{W_{AB}}{q}$$

$(W \cdot D) = q(\Delta V)$

Define 1 Volt? The electrostatic potential diff. B/w any two points in a electrostatic field is said to be 1 volt, when 1 Joule of work is done in moving a positive charge of 1C. from one point to another against F_e(field).

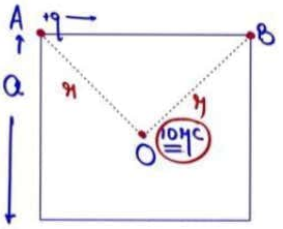
Find $W_{A \to B}$?

$W_{AB} = U_B - U_A$

$q(V_B - V_A)$

$V_B = V_A$

$W_{AB} = 0$



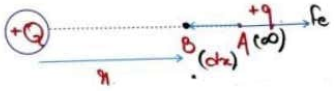
Electrostatic potential Due to a point charge.

$F = \frac{KQq}{r^2}$

$E = \frac{KQ}{r^2}$

$\int x^n = \frac{x^{n+1}}{n+1}$

$\int x^{-2} = \frac{x^{-2+1}}{-2+1} = -\frac{1}{x}$



$W = \int F_e \cdot dx \cos 180^\circ$

$= -\int_A^B F_e \cdot dx$

$W = -\int_\infty^r \frac{KQq}{x^2} \cdot dx$

$W = -KQq \int_\infty^r \frac{1}{x^2} dx$

$W = -KQq \left(-\frac{1}{x}\right)_\infty^r$

$W = KQq \left(\frac{1}{r} - \frac{1}{\infty}\right)$

$V = \frac{W}{q} = \frac{KQ}{r}$

$V = \frac{KQ}{r}$ volt

$V \propto \frac{1}{r}$

Dimension

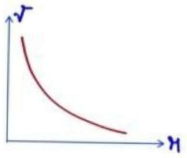
$V = \frac{W}{q} = \frac{ML^2T^{-2}}{AT}$

$V = [ML^2T^{-3}A^{-1}]$

$V = \frac{Kq}{r}$

$q \rightarrow +ve$, $V = +ve$

$q \rightarrow -ve$, $V = -ve$



position where potential is zero.

$V_1 - V_2 = 0$

$V_1 = V_2$

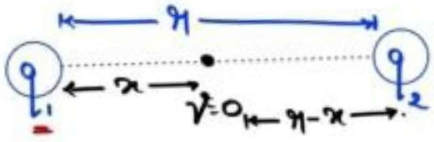
$\frac{Kq_1}{x} = \frac{Kq_2}{(r-x)}$

$\frac{r-x}{x} = \frac{q_2}{q_1}$

$q_1(r-x) = q = x$

$q_1 r - q_1 x = q = x$

$\frac{q_1 r}{(q_1+q_2)} = x$

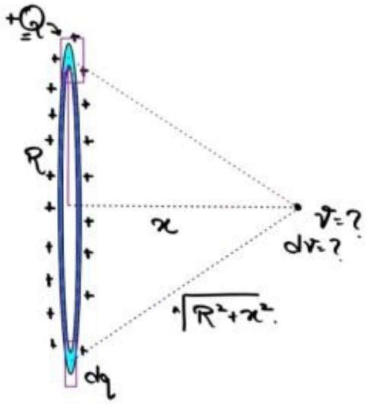


Electric potential Due to Ring

$\int_0^V dV = \frac{kx\,dq}{\sqrt{R^2+x^2}}$

$V = \frac{K}{\sqrt{R^2+x^2}} \int dq$

$V_{Ring} = \frac{KQ}{\sqrt{R^2+x^2}}$



Relationship Between Electric field and Electrostatic potential

$\Delta V = \frac{\Delta U}{q}$

$W = -\Delta U$

$V_B - V_A = -\frac{W_{A \rightarrow B}}{q}$

$V_B - V_A = -\int \frac{F \cdot dx}{q}$

$V_B - V_A = -\int q \frac{E \cdot dx}{q}$

$V_B - V_A = -\int E \cdot dx$

$\int dV = -\int E \cdot dx$

$V_B - V_A = -E \int dl$

$\Delta V = -E \cdot l$

Negative sign indicates thats if we move in the Direction $\vec{E}$ then potential Decrease.

$|dV| = E|dr|$

$E = \frac{dV}{dx}$, slope $\left(\frac{dy}{dx}\right)$ of potential vs Disp gives $|\vec{E}|$.

$dV = \int E dx$ ($\int y \cdot dx$ = Area under graph)

Area under $\vec{E}$ Vs $\vec{r}$ gives ΔV

$|\vec{E}| = \frac{-|dV|}{dx} = -$ (Potential Gradient)

The magnitude of electric field is given by change in magnitude of potential per unit Displacement normal to Equipotential surface at that point is called potential gradient

Co-ordinate system (x, y, z)

$V_B - V_A = -\int \vec{E} \cdot d\vec{x}$

$\vec{E} = E_x\hat{i} + E_y\hat{j} + E_z\hat{k}$

$d\vec{x} = dx\hat{i} + dy\hat{j} + dz\hat{k}$

$$V_B - V_A = -\left(\int_{x_1}^{x_2} E_x dx + \int_{y_1}^{y_2} E_y dy + \int_{y_1}^{z_2} E_z dz\right)$$

V = given $\vec{E}$ find.

$$E = -\frac{\partial V}{\partial r} = -\left(\frac{dv}{\partial x}\hat{i} + \frac{dv}{\partial y}\hat{j} + \frac{dv}{\partial z}\hat{k}\right)$$

$\vec{E}$ → given V = ?

POTENTIAL ENERGY

Work done = ΔU

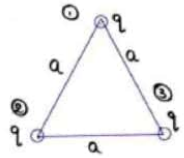
$$\int dw = -\int_A^B F \cdot dr = -\int_\infty^B \frac{Kq_1q_2}{r^2} \cdot dr$$

$$= -Kq_1q_2\left(\int r^{-2} dr\right)$$

$$= -Kq_1q_2\left(\frac{r^{-2+1}}{-2+1}\right)_{A=\infty}^{B=r}$$

$$U = \frac{Kq_1q_2}{r}$$

$$(\text{Workdone})_{ext} = U_{ext} = \frac{Kq_1q_2}{r}$$



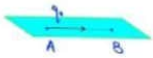
U_{21}

$U_{system} = U_{12} + U_{23} + U_{31}$

$$= \frac{Kq \times q}{} + \frac{Kq \times q}{} + \frac{Kq \times q}{}$$

EQUIPOTENTIAL SURFACES

- That surface at every point of which electric potential is same is called electrostatic potential.
- No work is done in moving a test change in equipotential surface.

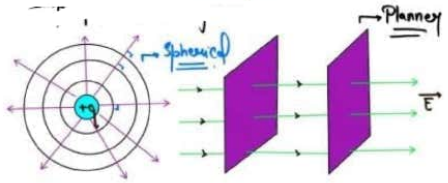


$w = q \cdot (\Delta V)$ → zero

$w = 0$

- Two equipotential through a point is normal to electric field.
- Two equipotential surface will never intersect each other.

→ some important equipotential surface.



Electric Dipole

Potential Due to electric Dipole.

$$V = -\frac{Kq}{r+l}, \quad V = \frac{Kq}{r-l}$$

$$V_{net} = \frac{Kq}{(r-l)} - \frac{Kq}{(r+l)}$$

$$V_{net} = Kq\left(\frac{1}{r-l} - \frac{1}{r+l}\right)$$

$$V_{net} = Kq\left(\frac{r+l-r+l}{r^2-l^2}\right)$$

$$V_{net} = \frac{Kq \times 2l}{r^2-l^2}$$

$$V_{net} = \frac{K|P|}{r^2-l^2}$$

$l << r$

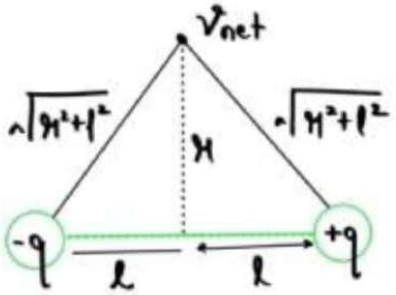
$$V_{at\ Axil} = \frac{Kp}{r^2}$$

$$V \propto \frac{1}{r^2}$$

Equitorial Position :-

$$V_{net} = \frac{-Kq}{\sqrt{r^2+l^2}} + \frac{Kq}{\sqrt{r^2+l^2}}$$

$V_{net} = 0$

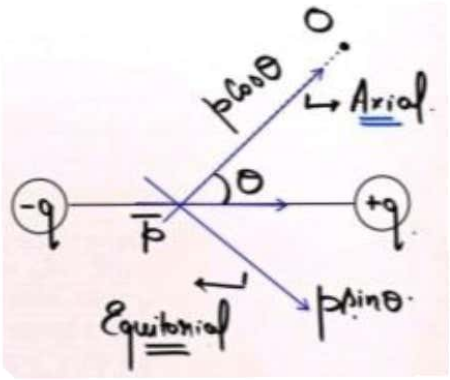


General point at $\vec{r}$ · θ

$$V = \frac{Kp\cos\theta}{r^2}$$

$\theta = 0°$, $V = \frac{Kp}{r^2}$ (Axial)

$\theta = 90°$, $V = \frac{Kp\cos 90°}{r^2} = 0$



POTENTIAL ENERGY AND TORQUE DUE TO DIPOLE

Torque :- $W = \int f \cdot dx$

$W = \int \tau \cdot d\theta$

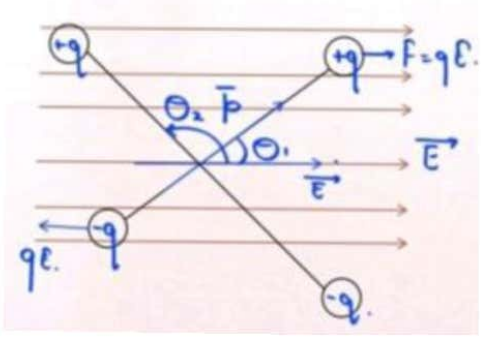
$W = \int_{\theta_1}^{\theta_2} PE \sin\theta \cdot d\theta$

$W = -PE(\cos\theta_2 - \cos\theta_1)$

$U = -PE\cos\theta$

$\theta = 0^\circ$ stable Eqⁿ

$\theta = 180^\circ$ Unstable Eqⁿ



Electrostatic potential Energy of a system of charges

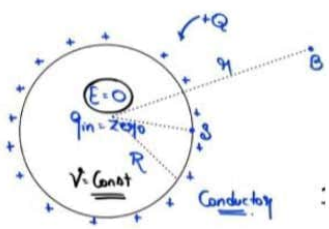
- Electrostatic potential Energy of a system of point charge is the total amount of work done, in bringing the various charge to their respective position.
- for a point charge.

ELECTROSTATICS OF CONDUCTORS

(1) Electrostatic field is zero inside a conductor.

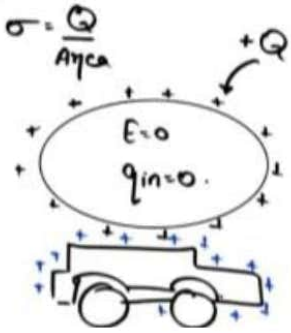
(2) At the surface of a charged conductor, electrostatic field must be normal to the surface at every point. if E were not normal to the surface, it would here some non-zero components along the surface. Free charges on the surface of the conductor would then experince force and move. In the static situtiion therepore E should have no tangential component.

(3) There is no excess charge at any point inside the conductor than the static situtiion, and any excess charge. must reside at the surface.



Electrostatic shielding

Electrostatic shielding/screening is the phenomenon of protecting a certain region of space from external electric field. we know that the electric field inside a conductor is zero. Therepore to protect delicate instruments from external electric fields, we enclose them in hollow conductors. such hollow conductors are called faraday cages. They need not be earthed.



Capacitor :-

✓ Provide large amount of energy at once.

$q \propto V$

$q = CV$

Energy store in the form of

- Capicitance is independent of charge and potential.

✡ Unit of C is Farad

✡ Dimension $C = \frac{Q}{V} = \frac{AT}{\frac{ML^2T^{-2}}{AT}}$

$C = M^{-1}L^{-2}A^2T^4$

Practical Unit of capacitance is $\mu F = 10^{-6} F$

Capacitance for spherical Capacitor

$C = \frac{Q}{V} = \frac{Q}{\frac{KQ}{r}} = \frac{1}{\frac{1}{4\pi\varepsilon_0 r}}$

$C = 4\pi\varepsilon_0 r$

The value of C Depends only on shape and size of conductor

Parallel plate capacitor

A (Area of plate)

$V = \int E \cdot dr$

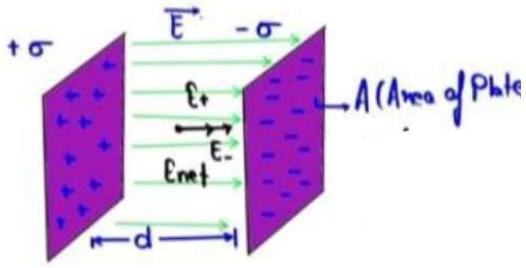
$V = \frac{\sigma}{\varepsilon_0} \int dr$

$V = \frac{\sigma d}{\varepsilon_0}$

$E_{net} = E + E = 2E$

$E_{net} = \frac{2 \times \sigma}{2\varepsilon_0}$

$E = \frac{\sigma}{\varepsilon_0}$



$C = \frac{Q}{V}$

$C = \frac{Q}{\frac{\sigma \times d}{\varepsilon_0}} = \frac{\varepsilon_0 \times Q}{\frac{Q}{A} \times d}$

$\sigma = \frac{Q}{A}$

$C = \frac{\varepsilon_0 A}{d}$

Energy stored

$dw = \int V \cdot dq$

$V = \frac{q}{C}$ $dw = \int \frac{q}{C} \cdot dq$

$w = \frac{q^2}{2C} = U$

Energy stored in Parallel Plate Capicitor

$U = \frac{q^2}{2C}$

$U = \frac{1}{2} \times \frac{q}{C} \times q$

$U = \frac{(CV)^2}{2C} = \frac{C^2 V^2}{2C}$

$U = \frac{qV}{2}$

$U = \frac{CV^2}{2}$

COMBINATION OF CAPACITOR

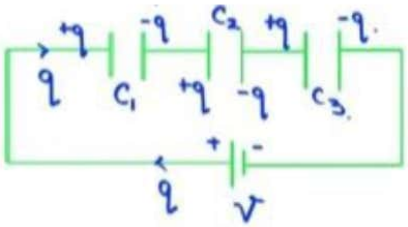
Series combination -

$V_1 = \frac{q}{C_1}$, $V_2 = \frac{q}{C_2}$, $V_3 = \frac{q}{C_3}$

$V_{net} = V_1 + V_2 + V_3$

$\frac{q}{C_{eq}} = \frac{q}{C_1} + \frac{q}{C_2} + \frac{q}{C_3}$

$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$

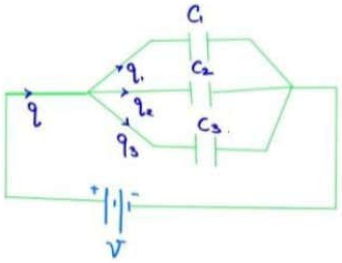


Parallel Combination -

$q_1 = C_1 V = q_2 = C_2 V$, $q_3 = C_3 V$

$q_{net} = C_{eq} \times V = C_1 V + C_2 V + C_3 V$

$C_{eq} = C_1 + C_2 + C_3$



Dielectric :-

$E_{net} = E_{ext} - E_{ind}$

$E_{net} = \frac{E_{ext}}{K}$

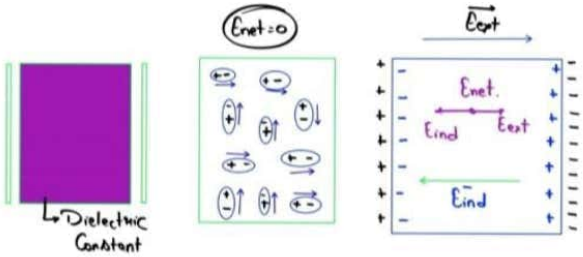
$\frac{E_{ext}}{K} = E_{ext} - E_{ind}$

$E_{ind} = E_{ext} - \frac{E_{ext}}{K}$

$\frac{\sigma i}{\varepsilon_0} = \frac{\sigma_0}{\varepsilon_0} \left(1 - \frac{1}{K}\right)$

$\sigma i = \sigma_0 \left(1 - \frac{1}{K}\right)$

$E = \frac{\sigma}{K\varepsilon_0}$



$C_0 = \frac{\varepsilon_0 A}{d}$

$C' = \frac{Q}{V} = \frac{Q}{E \cdot d} = \frac{Q}{\frac{\sigma}{K\varepsilon_0} \times d}$

$C' = \frac{K \times \varepsilon_0 A}{\frac{Q}{A} \times d}$

$C' = \frac{K\varepsilon_0 A}{d}$

$C' = KC_0$

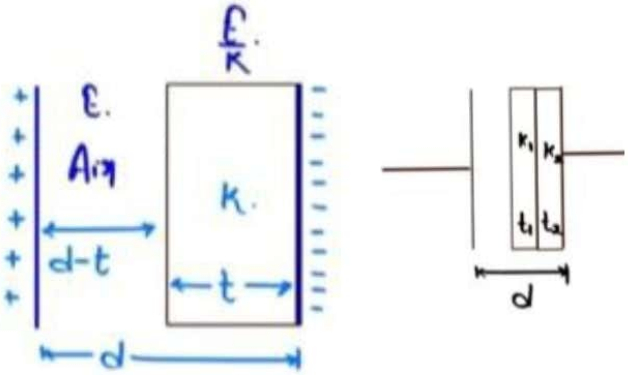
$$V = \varepsilon(d-t) + \frac{\varepsilon}{K} \times t$$

$$V = \varepsilon\left(d - t + \frac{t}{K}\right) = \frac{\sigma}{\varepsilon_0}\left(d - t + \frac{t}{K}\right)$$

$$C = \frac{Q}{V} = \frac{\cancel{\sigma}}{\frac{\cancel{\sigma}}{A\varepsilon_0}\left(d - t + \frac{t}{K}\right)}$$

$$C' = \frac{\varepsilon_0 A}{d - t + \frac{t}{K}}$$

@ $$C = \frac{\varepsilon_0 A}{d - t_1 - t_2 + \frac{t_1}{K_1} + \frac{t_2}{K_2}}$$



✡ Dielectrics are insulating materials which transmit electric effects without actually conducting electricity.

DIELECTRICS
- POLAR
- NON POLAR

NON-POLAR DIELECTRIC

- The molecules in which the centre of positive charge coincide with the centre of negative charge are called non-polar molecules.
- such molecules are symmetric in shape.

EX - O_2, CO_2, CH_4 etc